

Day 7 — Historical Context — Nārāyaṇa and Predecessors

Module: Magic Squares — Bhadragāṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: place *gaṇita-kaumudī* in 14th c. indian math.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *bhadra-gaṇita* (IAST: bhadra-gaṇita; lit. ““auspicious arithmetic” — the Indian term for magic squares”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Nārāyaṇa Paṇḍita (b. 1340 CE).** Mathematician of the Mādhava school period. His *Gaṇita-kaumudī* (“Moonlight of Computation”) is a major mathematical compendium. Book 14 (*Bhadra-gaṇita*) is dedicated to magic squares.
2. **Earlier Indian references.** Garga Saṁhitā (~1st c. BCE) mentions magic squares for ritual. The Khajuraho inscription (~10th c. CE) predates Nārāyaṇa by 4 centuries.
3. **Pre-Nārāyaṇa global timeline.**
 - Lo Shu legend (Chinese, attested ~5th–10th c. BCE): a 3×3 magic square.
 - Ibn Khaldūn and earlier Arab mathematicians: 9th–14th c. CE work on magic squares.
 - Khajuraho (~10th c. CE): pan-diagonal 4×4 .
 - Nārāyaṇa (1356 CE): systematic algorithms for any odd n .
 - Dürer (1514 CE): Melencolia I.
4. **Source.** Kim Plofker, *Mathematics in India* (Princeton, 2009), ch. 5 — best modern academic overview.

5. **Critical-thinking close.** “*Magic squares appear independently in many cultures. What does this tell us about (a) mathematical universality and (b) cultural transmission?*”
Nārāyaṇa’s contribution: ALGORITHMS for any odd n — a systematisation.
Khajuraho’s contribution: ART + math at temple scale.”

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Historical Context — Nārāyaṇa and Predecessors.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).